THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS TWO-CHAPTER TEST SET C

Chapters 1 & 2: **Patterns in Mathematics, Lines and Angles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The next number in the powers of 2 sequence $1, 2, 4, 8, 16, \ldots$ is: 1

(a) 36 **(b)** 24 **(c)** 48 **(d)** 32

2. The 10th triangular number is: 1

(a) 66 **(b)** 50 **(c)** 55 **(d)** 45

3. The sum of the first 9 odd numbers is: 1

(a) 81 **(b)** 72 **(c)** 90 **(d)** 64

4. In the Virahānka (Fibonacci) sequence 1, 2, 3, 5, 8, 13, … each term after the first two is: 1

(a) the difference of the previous two **(b)** the product of the previous two

(c) double the previous term **(d)** the sum of the two terms before it

5. A complete graph with 5 points has how many lines? 1

(a) 15 **(b)** 5 **(c)** 20 **(d)** 10

6. A ray has: 1

(a) one starting point and extends endlessly in one direction

(b) no starting point

(c) two end points

(d) a fixed length

7. An angle is formed by: 1

(a) two line segments of equal length **(b)** two parallel lines

(c) a circle **(d)** two rays with a common starting point

8. An angle of 90° is called a: 1

(a) acute angle **(b)** obtuse angle **(c)** reflex angle **(d)** right angle

9. An angle of 250° is a: 1

(a) obtuse angle **(b)** right angle **(c)** reflex angle **(d)** acute angle

10. The hour hand of a clock turns through how many degrees in one hour? 1

(a) 30° **(b)** 60° **(c)** 15° **(d)** 45°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. How many lines are there in a complete graph of 6 points? 2

12. Name two uses of patterns in everyday life. 2

13. What is a reflex angle? Give an example measure. 2

14. Write the number of right angles in a full turn. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How can a picture help us understand a number pattern? Illustrate with squares. 3

16. A wheel turns through 3 right angles. How many degrees is this? Is it acute, obtuse or reflex? 3

17. Write the Virahānka numbers up to 55. Which term is the sum of 13 and 21? 3

OR

Which is greater: the 7th square number or the 5th cube number? Show working. 3

18. Write the number of degrees made by a turn of 1/4, 1/2 and 3/4 of a circle. 3

OR

What are intersecting lines? How many points do they have in common? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A pattern: Figure 1 has 3 matchsticks, Figure 2 has 5, Figure 3 has 7. 5

(i) What is the rule?
(ii) Matchsticks in Figure 10?
(iii) Which figure uses 41?
(iv) Which sequence is this?

OR

(i) Show that 1+3+5+7+9+11 = 36 with a picture. 5
(ii) What is the sum of the first 20 odd numbers?
(iii) Is 169 the sum of some odd numbers starting from 1? How many?

20. (i) What are collinear points? 5
(ii) How many lines can pass through 2 points?
(iii) How many through 3 non-collinear points taken in pairs?

OR

(i) Find the angle through which the hour hand of a clock turns from 3 p.m. to 7 p.m. 5
(ii) the minute hand from 3:00 to 3:40.

END OF QUESTION PAPER — Total: 40 Marks

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